

DANIEL GARCIA RUANO

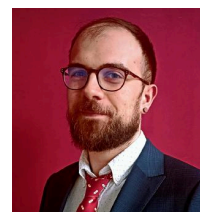
Geneticist and Bioinformatics Scientist

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[GitHub](#)

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PROFESSIONAL SUMMARY

Bioinformatician and geneticist with 7+ years of research experience spanning transcriptomics, machine learning for RNA biology, and experimental genetics. Current work focuses on **ML interpretability for lncRNA/mRNA classification** and **reproducible Snakemake pipelines for large-scale transcriptomic analyses**, with two manuscripts under review. Doctoral work applied whole-genome sequencing and variant-calling to experimental evolution, yielding three first-author peer-reviewed publications. I bring a rare combination of wet-lab and computational expertise, a consistent publication record across diverse biological questions, and a commitment to open-source and FAIR research practices.

EDUCATION

PhD in Cell Biology – "Revealing the link between cell volume and evolution" 2023

University of Rennes 1 / IGDR CNRS UMR6290

Master's Degree in Clinical Cancer Research 2019

University of Salamanca / Instituto de Biología Funcional y Genómica

Bachelor's Degree in Biotechnology 2018

University of Salamanca / Instituto de Biología Funcional y Genómica

PROFESSIONAL EXPERIENCE

Bioinformatics Scientist (Temporary Associate Professor) 2024 – Present

University of Bordeaux, Computational Biology & Bioinformatics Team (IBGC, UMR 5095)

- **Developed an uncertainty-aware benchmarking framework** for mRNA/lncRNA classification tools, revealing systematic tool disagreement for ~45% of transcripts and demonstrating the discriminative value of transposable elements and non-B DNA features ; first-author preprint on bioRxiv (2026).
- **Co-designed an interpretable deep-learning classifier** (multimodal: sequence + genomic context) for lncRNA identification, integrating transposable element content and non-B DNA motifs; manuscript under review at *Bioinformatics* (ECCB 2026).
- **Built end-to-end Snakemake pipelines** for (1) survival screening across 32 cancer types (in collaboration) and (2) benchmarking of computational transcript classification tools; all code versioned on GitHub.
- **Contributed to pydna and OpenCloning**, an open-source Python library and web app for molecular cloning design, focused on FAIR principles.
- **Supervised one Master's student** on transcript classification benchmarking.

Postdoctoral Fellow (Senior Genetics Scientist) 2023 – 2024

University of Bordeaux, SyntheCell Team

- **Applied RNA-seq and ChIP-qPCR** to dissect molecular basis of stress adaptation in fission yeast; manuscript in preparation.
- **Developed a CRISPR/Cas9 genome-editing strategy** to generate heterothallic strains from natural isolates for QTL mapping; published in *Yeast* (2024).

PhD Candidate (Junior Genetics Scientist) 2019 – 2023

University of Rennes / IGDR CNRS UMR6290

- **Implemented and extended the eVOLVER continuous culture platform** for long-term experimental evolution, delivering new **software tools for experiment control and data analysis**; published in *Open Biology* (2023).
- **Applied whole-genome sequencing and variant-calling** (GATK, FreeBayes) to track genomic adaptation across 16 evolving populations under 5 stress conditions.
- **Optimised fluorescence exclusion microscopy (FXm)** for high-throughput yeast cell volume measurement and developed an image analysis interface; published in *Journal of Cell Science* (2022).
- **Uncovered a stepwise adaptation mechanism**: SNV in *Toe2* transcription factor followed by a 10-copy CNV of target gene *SPBC36.01c*.

LANGUAGES AND CORE COMPETENCES

Programming: Python (pandas, scikit-learn, seaborn), R (featureCounts, DESeq2, ggplot2), Bash.

Workflow / HPC: Snakemake, Slurm/HPC, Git/GitHub, Conda

Omics & Analysis: RNA-seq (STAR, HISAT2, Salmon, htseq, Trinity), variant calling (GATK, FreeBayes)

ML / AI: Model interpretability, deep learning for sequence classification, benchmarking

Wet Lab: Fission yeast genetics, CRISPR/Cas9, ChIP-qPCR, fluorescence microscopy, microfluidics, flow cytometry

Languages: Spanish (native), English (C1, Cambridge-certified), French (professional proficiency)

PUBLICATIONS

- **García-Ruano D**, Georges M, Mohanty S, Makova K, Chalopin D and Nikolski M. Uncertainty-aware benchmarking reveals ambiguous transcripts in mRNA-lncRNA classification. *bioRxiv* (2026). [Preprint, first author] [\[DOI\]](#)
- Georges M, **García-Ruano D**, Mohanty S, Makova K, Chalopin D and Nikolski M. Interpretable multimodal learning from sequence and genomic context for lncRNA classification. *Submitted to Bioinformatics (ECCB proceedings)* (2026). [Under review]
- Naji F*, Oterino-Sogo S*, Beltsung F, **García-Ruano D**, ... Nikolski M, Rezvani HR. Spatial and Bulk Transcriptomics Defines the Molecular Evolution of Cutaneous Squamous Cell Carcinoma and Reveals Stage-Specific Biomarkers of Clinical Relevance. *bioRxiv, under revision at Cell Death and Disease* (2026). [\[DOI\]](#)
- Venkova L, **García-Ruano D**, Jain A, Charvin G, Coudreuse D. High-Throughput Measurement of Single-Fission Yeast Cell Volume Using Fluorescence Exclusion. *Methods in Molecular Biology* (2025). [Book chapter] [\[DOI\]](#)
- **García-Ruano D**, Hsu I, Leray B, Billard B, Liti G, Coudreuse D. Engineering heterothallic strains in fission yeast. *Yeast* (2024). [\[DOI\]](#)
- **García-Ruano D***, Jain A*, Heins ZJ, Wong BG, Yimer Wolle E, Khalil AS, Coudreuse D. Long-term evolution of proliferating cells using the eVOLVER platform. *Open Biology* (2023). [\[DOI\]](#)
- **García-Ruano D**, Venkova L, Jain A, Ryan JC, Radhakrishnan Balasubramaniam V, Piel M, Coudreuse D. Fluorescence exclusion – a rapid, accurate and powerful method for measuring yeast cell volume. *Journal of Cell Science* (2022). [\[DOI\]](#)
- López-Goñi I, ... **García-Ruano D**, ... et al. #EUROMicroMOOC: using Twitter to share trends in Microbiology worldwide. *FEMS Microbiology Letters* (2019). [\[DOI\]](#)

Selected presentations:

- Society for Molecular Biology and Evolution (SMBE) 2026, Copenhagen; Poster
- Oncosphere International Conference 2023, Bordeaux; Selected flash poster
- Journées Scientifiques ED-BS Bretagne-Loire 2022; Talk
- EMBL Conference 'Molecular Mechanisms in Evolution and Ecology' 2020; Poster

TEACHING EXPERIENCE

Temporary Associate Professor (ATER), Université de Bordeaux

2024 – Present

Full-time position (384 h) across 5 teaching units: **cell biology, developmental biology**, experimental imaging, and scientific communication; bilingual (FR/EN).

Guest Lecturer, University of Bordeaux

2023

Invited lecture on cell division and the cell cycle.

Teaching Assistant, University of Rennes 1

2022

Eukaryotic Genetics, EDUC Alliance Initiative (24 h).

SCIENCE COMMUNICATION & OUTREACH

- **Public engagement:** café talks, DNA extraction workshops for children, science fair activities on microscopy, molecular biology, and bioinformatics.
- **EUROMicroMOOC contributor:** co-authored publication in *FEMS Microbiology Letters* (2019).
- **SoyBiotec initiative:** YouTube channel and social media covering genetics, cancer research, and science news for general audiences.

FUNDING & AWARDS

- 4th-year PhD extension grant; Fondation ARC pour la Recherche contre le Cancer (2022)
- Science communication prize; Fundación Antama (2019)
- Master's scholarship; Institute of Biomedicine of Salamanca, IBSAL (2019)